

FDE-Associate Foundation — Advanced MCQ Exam

Based on “What AI Actually Is: A Crash Course” from The AI Agent Factory / Panaversity. 35 scenario-based questions
• 19 single-answer • 8 two-answer • 8 table-based. Answers are shown immediately after each question.

Source reviewed: <https://agentfactory.panaversity.org/docs/what-ai-actually-is-crash-course>

Question 1

A customer asks an internal support agent to answer questions about a proprietary product launched last month. The base model gives polished answers, but several launch-specific details are wrong. The customer says, “The model must have poor reasoning.” What is the most appropriate first diagnosis based on the course's mental model?

- A. The model is failing because inference permanently changes its weights after each answer.
- B. The model lacks the relevant information in its frozen weights, so it is predicting a plausible continuation rather than retrieving a verified product fact.
- C. The model is guaranteed to know every fact before its knowledge cutoff, so the issue must be temperature.
- D. The model has private company information internally but is refusing to reveal it.

Answer: B

Question 2

An FDE is building a workflow where a model must answer “What is the current status of ticket 84721?” The ticket system is updated hourly. Which architecture most directly addresses the stale-knowledge problem described in the course?

- A. Fine-tune the model after every ticket update.
- B. Increase temperature so the model explores more possible answers.
- C. Give the model an appropriate connector/tool to retrieve the current ticket, then place the result into context before answering.
- D. Add more examples of old tickets to the system prompt.

Answer: C

Question 3

A model is corrected in a chat: “The deployment date is Friday, not Thursday.” It immediately uses Friday correctly for the rest of the conversation, but starts a new chat and repeats Thursday. What best explains this?

- A. The model partially updated its weights but forgot the update later.
- B. The correction entered the current context, but inference did not change the frozen weights.
- C. The new chat has a higher temperature by default.
- D. The knowledge cutoff moved backward.

Answer: B

Question 4

A customer wants an AI assistant to use yesterday's private sales spreadsheet. The base model has no connector and the spreadsheet was never part of training. Which statement is most accurate?

- A. The model can infer the spreadsheet contents from its general training.
- B. The model has the data but its safety layer blocks it.
- C. The model can use the spreadsheet only if its contents are placed into the context window, directly or through a tool.
- D. The model will learn the spreadsheet permanently after reading it once.

Answer: C

Question 5

An FDE notices that lowering temperature makes an extraction workflow more repeatable, while raising it produces more varied outputs. What mechanism is being controlled?

- A. The size of the model's frozen weights.
- B. How broadly the system samples from the spread of likely next tokens.
- C. Whether the model can access the context window.
- D. Whether training or inference is running.

Answer: B

Question 6

A team claims: "The model sounds certain, so it probably has strong evidence." Which response best reflects the course?

- A. Confidence is a learned style and is not a truth signal; verification must be external or human.
- B. Confidence directly measures the probability that the statement is true.
- C. Confidence is only unreliable when tools are enabled.
- D. Confidence becomes a truth signal when temperature is low.

Answer: A

Question 7

An agent successfully writes complex production code but repeatedly fails a simple three-step logic puzzle. The customer concludes the model is "inconsistent and therefore unusable." What is the more useful engineering interpretation?

- A. The model's capability is jagged; difficult-looking tasks do not reliably predict performance on neighboring tasks.
- B. The model must be secretly changing its weights between prompts.
- C. The logic puzzle is necessarily harder computationally than the code.
- D. The failure proves that the model has no useful generalization.

Answer: A

Question 8

A 200-page contract is relevant to a question about one indemnity clause. A team instead pastes a 300-word summary and receives an answer that misses a critical exception in the original. Which principle best explains the failure?

- A. The model's weights are too small to understand legal language.
- B. Only information present in the context window can directly inform the answer; the summary omitted information that mattered.
- C. The model must be retrained on the contract.
- D. Temperature should be increased to make the model more attentive.

Answer: B

Question 9

A long-running chat becomes slower and increasingly expensive on an API. The application sends the entire conversation transcript with every turn. What is the primary mechanical cause?

- A. The model's weights become larger as the transcript grows.
- B. Each turn reprocesses the growing transcript as context, increasing token work.
- C. The model begins training during long conversations.
- D. The knowledge cutoff is recalculated on every turn.

Answer: B

Question 10

An FDE needs a workflow to access a customer's Drive and Gmail without writing separate bespoke integrations for every agent implementation. Which concept from the course is most relevant?

- A. Temperature.
- B. MCP as a shared standard through which connectors expose permitted app access.
- C. Knowledge cutoff.
- D. RLHF as a data transport protocol.

Answer: B

Question 11

A pure language model is asked to calculate a complex financial projection. It produces a plausible number. The FDE instead wants a verifiable calculation. Which design best matches the course?

- A. Ask the model to sound more confident.
- B. Use code execution so the model can generate a program, execute it, and receive the real result.
- C. Increase temperature and sample ten answers.
- D. Put “be mathematically correct” in the system prompt and trust the output.

Answer: B

Question 12

A model is asked about an obscure self-published book with almost no online discussion and invents a detailed plot. Why is this failure especially predictable from the course?

- A. Rare topics have weaker learned continuations, so the model may produce a plausible blend without an internal truth checker.
- B. Rare topics automatically trigger a lower context window.
- C. The model cannot generate text about books unless fine-tuned.
- D. The model only hallucinates when temperature is exactly zero.

Answer: A

Question 13

A team wants a model to answer instructions rather than simply continue arbitrary text. Which training stage is primarily responsible for teaching that behavior?

- A. Quantization.
- B. Instruction tuning.
- C. Tokenization.
- D. Inference.

Answer: B

Question 14

A customer asks why an AI product cannot simply “learn from each conversation” while preserving reliable behavior for millions of users. Which combination from the course best explains the deliberate freeze?

- A. Cost, safety/testing, and consistency.
- B. Temperature, tokenization, and image patches.
- C. Connectors, projects, and artifacts.
- D. Search, memory, and model routing.

Answer: A

Question 15

An FDE is choosing between a fast low-cost model and a more expensive high-depth model. The task is a routine bulk classification job with low consequences. Which decision principle is most consistent with the course's model ladder guidance?

- A. Always choose the largest model because larger models eliminate all errors.
- B. Prefer the expensive model because cost is irrelevant at inference.
- C. Use the smaller fast model when the task is high-volume and low-depth, and reserve deeper models for tasks that need them.
- D. Choose randomly because model capability is identical.

Answer: C

Question 16

A team enables a high-effort reasoning mode for every request, including simple formatting tasks. Their costs and latency rise with little visible benefit. What is the most direct explanation?

- A. Reasoning generates additional working tokens, which consume time and budget.
- B. Reasoning permanently increases the model's parameter count.

- C. Reasoning disables tools.
- D. Reasoning expands the knowledge cutoff.

Answer: A

Question 17

An assistant uses a memory feature to recall a user's preferred report format across new chats. According to the course's product-level explanation, what is mechanically happening?

- A. The model's weights are rewritten after every conversation.
- B. The product stores a note and re-inserts relevant text into the context window.
- C. The model reads the user's mind from hidden parameters.
- D. The knowledge cutoff is extended for that user.

Answer: B

Question 18

A customer asks whether a polished answer with three citations can be trusted because “the model would not cite sources it had not seen.” What should the FDE do first?

- A. Accept the citations because citation formatting is evidence of truth.
- B. Verify that the cited works actually exist and support the claims; the model has no separate internal truth-checker.
- C. Increase temperature to encourage more honest citations.
- D. Fine-tune on the customer's citation style before checking.

Answer: B

Question 19

A team wants to reduce failures on a high-impact workflow. They have already tested the model on difficult benchmark examples, where it performs well. The course suggests a different verification strategy. Which is it?

- A. Verify only the hardest examples because easy examples are inherently safe.
- B. Verify across the jagged boundary, including easy-looking tasks that the model may fail unexpectedly.
- C. Stop testing once the model passes a difficult benchmark.
- D. Use confidence of wording as a proxy for reliability.

Answer: B

Question 20

A customer says an agent “remembers” a detail from message 20 of the same chat. Which TWO mechanisms are consistent with the course?

- ☐ A. The model permanently stored the fact in its weights.
- ☐ B. The transcript is stored by the product and replayed into context on later turns.
- ☐ C. The model retrained itself after message 20.
- ☐ D. The relevant message remains available in the current context window.
- ☐ E. The knowledge cutoff was dynamically updated.

Correct answers: B and D

Question 21

An FDE is diagnosing a hallucinated answer about a niche customer policy. Which TWO actions most directly implement the course's “you are the missing checker” principle?

- ☐ A. Verify the policy against the authoritative source.
- ☐ B. Treat a confident tone as sufficient evidence.
- ☐ C. Use an appropriate retrieval/connector tool to place current policy text into context.
- ☐ D. Increase temperature until the answer changes.
- ☐ E. Assume the model's first answer is its highest-confidence truth.

Correct answers: A and C

Question 22

A multilingual deployment has unexpectedly high token usage and shorter effective context than the English pilot. Which TWO explanations are supported by the course?

- ☐ A. Non-English scripts can tokenize into more tokens per word.
- ☐ B. The model automatically doubles its parameter count for non-English text.
- ☐ C. More tokens can increase cost and fill the context window faster.
- ☐ D. Token counts are based only on characters, so language cannot matter.
- ☐ E. The knowledge cutoff becomes shorter in non-English languages.

Correct answers: A and C

Question 23

A customer wants an agent to take real actions in a CRM rather than merely recommend them. Which TWO architectural elements are central to the course's definition of an agent?

- ☐ A. A predictor that can choose an action/tool call.
- ☐ B. A separate consciousness module.
- ☐ C. A tool that executes the action and returns an observation/result.
- ☐ D. A larger training dataset is mandatory at runtime.
- ☐ E. A higher temperature setting.

Correct answers: A and C

Question 24

A team is deciding where to place instructions in a Claude-like product. Which TWO statements match the appendix's scope guidance?

- ☐ A. Account-wide instructions should contain only guidance true for essentially every conversation.
- ☐ B. Topic-specific instructions are best placed in every account-wide instruction field.
- ☐ C. Project instructions can apply only to chats inside that project.
- ☐ D. Project knowledge changes the model's frozen weights.
- ☐ E. A project is unrelated to context management.

Correct answers: A and C

Question 25

A workflow needs current information and a multi-source investigative report. Which TWO tool choices align with the appendix?

- ☐ A. Use web search when you need a small number of current facts.
- ☐ B. Use Research when you need a multi-step search process producing a cited report.
- ☐ C. Avoid tools because the frozen weights are always current.
- ☐ D. Use higher temperature instead of search.
- ☐ E. Use memory as a substitute for current external facts.

Correct answers: A and B

Question 26

A reasoning model produces a better answer after generating intermediate working, but a critical factual error remains. Which TWO conclusions follow from the course?

- ☐ A. More thinking can improve performance by putting useful working into context.
- ☐ B. Reasoning creates a separate truth-checking organ.
- ☐ C. Reasoning can still miss errors because the working is produced by the same prediction process.
- ☐ D. A longer chain guarantees factual correctness.
- ☐ E. The knowledge cutoff disappears during reasoning.

Correct answers: A and C

Question 27

A team is validating a model before production. Which TWO testing practices are explicitly supported by the course's discussion of jagged capability?

- ☐ A. Test both hard tasks and deceptively easy tasks.

- ☐ B. Assume success on a hard task guarantees success on easier neighboring tasks.
- ☐ C. Compare the same task across multiple models when practical.
- ☐ D. Test only tasks that appear frequently in benchmark datasets.
- ☐ E. Use confidence of prose to avoid manual verification.

Correct answers: A and C

Question 28

A customer support agent is missing current account information. The FDE tests four designs. Based only on the table, which design best addresses the stated need while preserving the model's frozen-weight architecture?

Design	Data source	Runtime behavior	Main limitation	Fit for current account status
A	Training corpus	Predict from weights	May be stale	Poor
B	Connector to CRM	Fetch current record into context	Requires granted permissions	Strong
C	Longer system prompt	Static policy text	Does not update automatically	Weak
D	Higher temperature	More varied sampling	Does not add facts	Poor

- A. The first row is the correct interpretation.
- B. The second row is the correct interpretation.
- C. The third row is the correct interpretation.
- D. The fourth row is the correct interpretation.

Answer: B

Question 29

An FDE investigates rising API cost in a long-running assistant. The customer wants the most direct explanation supported by the measured behavior. Which row identifies the best diagnosis?

Case	Transcript length	Input tokens/turn	Latency	Observed cost
A	10k → 20k	10k → 20k	Rising	Rising
B	10k → 10k	Flat	Rising	Flat
C	20k → 5k	Falling	Falling	Rising
D	5k → 5k	Flat	Flat	Rising

- A. The first row is the correct interpretation.
- B. The second row is the correct interpretation.
- C. The third row is the correct interpretation.
- D. The fourth row is the correct interpretation.

Answer: A

Question 30

A model is evaluated on five tasks before deployment. The team wants to avoid assuming that apparent difficulty predicts reliability. Which conclusion from the table most directly demonstrates the jagged frontier?

Task	Human-perceived difficulty	Model result	Frequency in training-like text	Verification need
A	Very hard	Excellent	High	Normal
B	Easy letter count	Poor	Low	High
C	Hard coding	Excellent	High	Normal
D	Easy logic puzzle	Poor	Low	High
E	Moderate summary	Excellent	High	Normal

- A. The first row is the correct interpretation.
- B. The second row is the correct interpretation.
- C. The third row is the correct interpretation.
- D. The fourth row is the correct interpretation.

Answer: B

Question 31

A product team compares ways to provide a model with customer-specific knowledge. The customer wants the information available at inference time without changing the model weights. Which option is the best fit?

Option	Changes weights?	Information available at inference?	Persistence mechanism	Primary risk
A	Yes	Yes	Fine-tuning	Model drift/cost
B	No	Yes	Context injection	Context limits
C	No	No	Training cutoff	Staleness
D	Yes	Yes	Per-chat learning	Consistency loss

- A. The first row is the correct interpretation.
- B. The second row is the correct interpretation.
- C. The third row is the correct interpretation.
- D. The fourth row is the correct interpretation.

Answer: B

Question 32

An FDE is selecting an execution strategy for four workloads. The goal is to match model effort to task requirements rather than maximizing model size. Which selection is most consistent with the course?

Workload	Depth	Volume	Latency sensitivity	Recommended approach
A	Low	Very high	High	Small/fast model
B	High	Low	Low	Deep/high-capability model
C	Simple formatting	Very high	High	Small/fast model
D	Complex architecture	Low	Low	Deep/high-capability model

- A. The first row is the correct interpretation.
- B. The second row is the correct interpretation.
- C. The third row is the correct interpretation.
- D. The fourth row is the correct interpretation.

Answer: A

Question 33

A team is deciding whether a tool is required. The table shows what the base predictor can know from frozen weights versus what the tool can supply at runtime. Which row correctly captures the role of a tool?

Need	Frozen weights	Tool result	Where result enters	Correct interpretation
A	Today's weather	May be stale/unknown	N/A	Tool can fetch current data
B	Old common fact	Often strong	N/A	Tool always required
C	Private file contents	Not inherently present	Context window	Tool can retrieve and expose it
D	Current database row	Not guaranteed current	Context window	Tool can fetch live state

- A. The first row is the correct interpretation.
- B. The second row is the correct interpretation.
- C. The third row is the correct interpretation.
- D. The fourth row is the correct interpretation.

Answer: A

Question 34

A customer reports that an assistant agrees with leading questions even when evidence points the other way. The FDE tests neutral and leading prompts. Which interpretation is most consistent with the table?

Prompt	Framing	Answer style	Evidence check	Interpretation
A	"Isn't X clearly correct?"	Strong agreement	Absent	Sycophancy risk
B	"Evaluate X and give strongest case on each side."	Balanced	Requested	Less cueing toward agreement
C	"You must agree with me."	Agreement	Absent	Expected cue-driven behavior

Prompt	Framing	Answer style	Evidence check	Interpretation
D	"Isn't X obviously true?"	Agreement	Absent	Confidence proves truth

- A. The first row is the correct interpretation.
- B. The second row is the correct interpretation.
- C. The third row is the correct interpretation.
- D. The fourth row is the correct interpretation.

Answer: A

Question 35

A team is building an agent that must inspect a file, calculate a value, and then update a system. Which row best represents the predict-act-observe loop described in the course?

Step	Model prediction	External action	Returned information	Next state
A	Choose file-read tool	Read file	File contents	Contents enter context
B	Guess final value	No execution	None	Guess remains unsupported
C	Choose calculator tool	Run calculation	Computed result	Result enters context
D	Choose update tool	Write CRM record	Success/failure	Observation can guide next action

- A. The first row is the correct interpretation.
- B. The second row is the correct interpretation.
- C. The third row is the correct interpretation.
- D. The fourth row is the correct interpretation.

Answer: A

Exam design note: Questions emphasize mechanism-level reasoning, troubleshooting, architecture choices, context/tool behavior, verification, jagged capability, and agent loops. No answer explanations are included, per the requested exam format.